

BONE MARROW ASPIRATE, BIOPSY & CYTOGENETICS
REPORT

Date: September 11, 2026

Patient Name:	Vikram Reddy	MRN / Patient ID:	CMC-HEM-781902
Age / Gender:	19 Yrs / Male	Blood Group & Rh:	O+ (O Rh Positive)
Clinical Diagnosis:	Idiopathic Severe Aplastic Anemia (SAA)	Referring Consultant:	Dr. Alok Srivastava, MD, FRACP

Morphological / Cytogenetic Marker	Observed Value	Normal Reference	Clinical Evaluation
Marrow Cellularity (Trepine)	< 10% (Hypocellular)	40% - 60%	Severe Marrow Aplasia
Myeloblast Percentage	1.2%	< 5.0%	Normal / No Excess Blasts
Trilineage Hematopoiesis	Markedly Depressed	Normal active trilineage	Marrow Failure State
Cytogenetic Karyotype (G-Banding)	46,XY (20 Metaphases)	Normal Diploid 46,XY	Normal Favorable Karyotype
FISH Monosomy 7 / 5q- Deletion	Negative / Normal Signals	Absence of MDS clones	No Malignant Clones

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- Bone marrow biopsy confirms marked hypocellularity with preservation of fat spaces, diagnostic of Severe Aplastic Anemia.
- Normal 46,XY karyotype rules out myelodysplastic syndrome (MDS) transformation or high-risk clonal evolution.
- Young age (< 20 years) indicates urgent allogeneic stem cell transplantation as first-line curative therapy.

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